LineageRank and PipeRCA-Bench:

Root-Cause Ranking in Data Pipelines under Partial Observability

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Abstract—
Modern data pipelines fail in ways that are easy to observe but difficult to diagnose. Existing lineage systems expose what failed and which assets depend on what, yet leave a gap between metadata availability and culprit prioritization. We formulate pipeline root-cause analysis (RCA) as a ranked retrieval problem and introduce PipeRCA-Bench, a reproducible benchmark of 450 labeled incidents spanning three pipeline families, six fault classes, and three lineage observability conditions. We propose four ranking methods: LineageRank-H, an interpretable weighted-sum heuristic; LineageRank-CP, a causal propagation variant that exploits directional evidence gradients along the lineage graph; LineageRank-BS, a partial-observability-aware method that multiplicatively amplifies scores for nodes with design-runtime lineage discordance; and LineageRank-L, a Random Forest learned ranker. LR-BS achieves Top-1 accuracy 0.8578—a 0.24 absolute improvement over LR-H and within 0.04 of the learned model—while remaining fully interpretable. Under the runtime-missing-root observability condition, LR-BS achieves perfect Top-1 1.000 by directly exploiting lineage-coverage gaps as root-cause evidence. A real public-data case study on NYC TLC taxi records with captured runtime lineage corroborates the findings. The full setup runs with open-source tools on laptop-scale hardware.

Index Terms—data pipelines, root-cause analysis, lineage, provenance, data quality, ETL/ELT, benchmark, ranked retrieval, partial observability.

I. INTRODUCTION

Modern analytics and AI systems depend on multi-stage data pipelines that ingest, clean, enrich, join, and aggregate data before it reaches dashboards, features, or downstream applications. In practice, failures are easy to observe but difficult to trace: a failing dashboard, a broken contract, a stale aggregate, or a suspicious metric may reflect a fault that originated several steps upstream. Foidl et al. [1] document that quality problems in practical pipelines are common and concentrated around cleaning, integration, and type issues, reinforcing that upstream diagnosis remains a persistent engineering challenge.

The open-source data ecosystem is now rich in operational metadata. OpenLineage and Marquez standardize runtime lineage capture [2]; the dbt transformation framework exposes tests, contracts, and freshness artifacts [3][4][5][6]; Airflow provides dataset-aware scheduling metadata [7]; and OpenMetadata surfaces incident workflows [8]. Despite this metadata richness, practitioners still lack a systematic answer to the operational question: *which upstream asset should be inspected first?*

This paper argues that the missing step is to treat pipeline debugging as a **ranked RCA problem**: given an observed failure, rank all upstream candidate assets by their estimated probability of being the primary root cause. Two observations motivate this framing. First, lineage alone is insufficient: runtime lineage may be incomplete due to instrumentation gaps, and OpenLineage itself has introduced static and code-derived lineage to address these blind spots [9][10]. Second, adjacent communities have shown that benchmark-driven RCA is a reusable and productive research direction; RCAEval [11] established a structured comparison substrate for microservice RCA, yet data pipelines lack a comparable open evaluation artifact.

We contribute:

1. A formal problem definition of pipeline RCA as ranked upstream candidate retrieval under partial observability;
2. PipeRCA-Bench, a benchmark with 450 labeled incidents spanning three pipeline families, six fault classes, and three observability conditions;
3. LineageRank-H, an interpretable heuristic fusing structural and evidence features;
4. LineageRank-CP, a novel causal propagation ranker exploiting directional evidence gradients along lineage paths;
5. LineageRank-BS, a partial-observability-aware heuristic that achieves near-learned-model accuracy by amplifying scores for design-runtime lineage discordance;
6. LineageRank-L, a lightweight Random Forest learned ranker with leave-one-pipeline-out cross-validation;
7. A leakage and ablation audit, bootstrap confidence intervals, and paired significance tests;

8. A real public-data case study with captured runtime lineage on official NYC TLC trip records.

II. PROBLEM FORMULATION

We focus on dataset/job-level RCA for batch ETL/ELT pipelines. Let G = (V, E) be a directed acyclic graph where V is the set of data assets (sources, staging models, marts) and E is the set of directed dependency edges (u, v) indicating u is upstream of v. We distinguish design-time edges E_d, derived from static pipeline definitions, from runtime edges E_r, captured from execution lineage events. The fused graph uses G_f = (V, E_d ∪ E_r).

> ***Definition 1 (Pipeline RCA Task).*** *Given (i) an observed failure at asset v_obs at time t, (ii) a set of evidence signals S mapping each node u ∈ V to observable quality, freshness, and anomaly indicators available up to time t, and (iii) the fused lineage graph G_f, the ranked RCA task is to produce a ranked list of candidate upstream assets C = ancestors(v_obs, G_f) ∪ {v_obs} such that candidates are ordered by their estimated probability of being the primary root cause.*

Evaluation uses the rank position of the known ground-truth root cause. Primary metrics are Top-k accuracy and Mean Reciprocal Rank (MRR). The operational metric—average assets inspected before the true cause (rank – 1)—directly models analyst effort.

The formulation excludes record-level blame assignment, automated remediation, proprietary observability systems as a hard dependency, GPU-heavy deep learning, and column-level RCA, keeping the task feasible, reproducible, and aligned with the OSS data engineering toolchain.

III. RELATED WORK

A. Provenance and Lineage Systems

Provenance capture and lineage systems provide the foundational infrastructure for this work but do not themselves define a benchmarked ranked RCA task for ETL/ELT incidents. OpenLineage and Marquez standardize runtime lineage collection in modern data stacks [2]. Chapman et al. [12] introduce fine-grained provenance for data science pipelines, enabling pipeline inspection queries but not ranked RCA evaluation. Schelter et al. [13] use provenance to screen ML data-preparation pipelines for correctness and fairness issues. Johns et al. [14] integrate provenance into clinical ETL quality dashboards. These systems support capture, screening, and tracing, but do not define a benchmarked ranked RCA task.

B. Data Pipeline Quality

Foidl et al. [1] characterize common quality problems in practical pipelines and find cleaning, integration, and type issues are especially prevalent. Vassiliadis et al. [15] study schema evolution at EDBT 2023, showing structural changes propagate through pipeline graphs. Barth et al. [16] study stale data in data lakes, providing empirical evidence that freshness degradation is a practically impactful failure mode. These works ground our fault taxonomy in empirical evidence.

C. Causal Inference for System RCA

Causal structure learning has been applied to microservice RCA with methods such as CIRCA (Li et al., 2022), which uses a Causal Bayesian Network to bridge observational data and interventional knowledge for microservice fault localization [17]. CausalRCA (Xin et al., 2023) automatically builds anomaly propagation graphs for microservice RCA using causal inference [18]. Our causal propagation variant (LR-CP) is conceptually motivated by this literature but applied to the data pipeline domain where lineage graph semantics, freshness metadata, and dbt-style artifacts replace microservice call graphs and metric timeseries.

D. Benchmark-Driven RCA in Adjacent Domains

In microservices, RCA has matured into a benchmarked literature. RCAEval [11] consolidates nine real-world datasets with 735 failure cases and an evaluation framework for comparing localization methods. Graph-free RCA methods (e.g., PRISM [19]) further advance this agenda, achieving 68% Top-1 on RCAEval. Our work draws on the benchmark-driven methodology of RCAEval but operates over fundamentally different graph semantics, evidence types, and failure modes. PC-PageRank—a graph-based approach evaluated within RCAEval—achieves only 9% Top-1 accuracy, illustrating that methods designed for microservice call graphs do not transfer directly to our setting.

E. Observability Tools

Modern data observability tools—including dbt's test and contract system [4][5], Airflow's dataset-aware scheduling [7], and OpenMetadata's incident workflows [8]—expose rich operational evidence signals. Our work combines these signals into a principled ranked RCA framework that is benchmarked and reproducible.

IV. THE LINEAGERANK FRAMEWORK

A. Evidence Graph Construction

LineageRank operates over a fused evidence graph combining runtime lineage edges E_r (captured via OpenLineage-compatible event emission) and design-time lineage edges E_d (derived from pipeline specifications or dbt manifests). For each incident with observed failure v_obs, the candidate set is C = ancestors(v_obs, G_f) ∪ {v_obs}.

B. Feature Extraction

For each candidate node u ∈ C, LineageRank computes a 16-dimensional feature vector across four groups. **Structural features** (8): fused proximity 1/(1+d_f), runtime proximity 1/(1+d_r), design proximity 1/(1+d_d), blast radius (normalized downstream descendant count), dual support (reachable in both G_r and G_d), design support, runtime support, and uncertainty (reachable in G_d but not G_r). **Observability feature** (1): blind-spot hint, set when a node is design-reachable but absent from runtime lineage—directly capturing partial observability. **Evidence features** (6): quality signal (local test failures and contract violations), failure propagation (weighted downstream test failures

in the impacted set), recent change, freshness severity, run anomaly, and contract violation indicator. **Prior feature** (1): fault prior, a lookup mapping fault type and node type to a probability, encoding domain knowledge (stale-source faults typically originate at source nodes; bad-join-key faults at join transformations).

C. Heuristic Variant (LineageRank-H)

LineageRank-H produces an interpretable ranking score as a weighted sum:

score_H(u) = 0.17 prox + 0.15 blast + 0.12 blind_spot + 0.12 prior + 0.11 design_sup + 0.10 fresh + 0.08 change + 0.08 quality + 0.08 prop + 0.07 dual + 0.06 anomaly - 0.04 uncertainty

where terms are as defined in the feature groups above. Weights were set by structured manual analysis of feature behavior across held-out pilot incidents. To assess robustness, we evaluate three perturbation variants: a uniform baseline (all features equal weight), a high-evidence variant (evidence and observability weights doubled, structural halved, then renormalized), and a high-structural variant (structural doubled, evidence halved). Top-1 accuracy is 0.3778 (uniform), 0.8378 (high-evidence), and 0.2822 (high-structural). The uniform baseline substantially underperforms the tuned weights, confirming the weights are informative. More strikingly, the high-evidence variant (0.8378) approaches LR-BS performance (0.8578), revealing that additive amplification of evidence and observability features is the dominant lever. This observation directly motivates the LR-BS design: rather than requiring manual re-tuning of all weights, LR-BS applies principled multiplicative amplification specifically to the blind-spot signal, achieving superior performance with a transparent mechanism.

D. Causal Propagation Variant (LineageRank-CP)

LineageRank-CP introduces a directional evidence gradient signal motivated by the observation that root causes tend to exhibit higher local evidence than their downstream descendants, since the fault originates at the root and signal attenuates as it propagates. For each candidate u with descendants D(u) in G_f, we define:

evidence_gradient(u) = max(0, ev(u) − avg_{v ∈ D(u)} ev(v))

where ev(u) = 0.50 × run_anomaly + 0.30 × recent_change + 0.20 × freshness_severity. The CP score is then:

*score_CP(u) = 0.28 proximity + 0.30 local_ev + 0.22 evidence_gradient + 0.10 failure_propagation + 0.10 * fault_prior*

This formulation is parameter-lean (five coefficients, manually set to reflect relative signal importance) and does not require the fault prior to be the dominant term.

E. Blind-Spot Boosted Variant (LineageRank-BS)

LineageRank-BS is motivated by a key empirical observation: when a root cause node fails to emit runtime lineage events (the runtime_missing_root condition), its outgoing edges are absent from the runtime graph. This creates a distinctive signature: the node is reachable in design lineage but absent from runtime lineage—the blind-spot indicator. Rather than treating this as a single additive feature (as in LR-H), LR-BS applies a multiplicative amplification:

score_BS(u) = base(u) × (1 + 2.5 × blind_spot_hint(u))

where base(u) = 0.25 × proximity + 0.35 × local_ev + 0.15 × failure_propagation + 0.15 × fault_prior + 0.10 × blast_radius. Under full observability (no missing edges), blind_spot_hint = 0 for all nodes and BS reduces to the base score. Under runtime_missing_root, BS uniquely identifies the root cause through the amplification factor, achieving Top-1 accuracy of 1.000 in this condition.

F. Learned Variant (LineageRank-L)

LineageRank-L trains a Random Forest classifier [20] over the same 16-dimensional feature vector, treating binary label (node == root cause) as the classification target. To guard against data leakage, we use leave-one-pipeline-out cross-validation: training on incidents from two pipeline families and evaluating on the third. Feature importances are averaged across held-out runs.

V. PIPERCA-BENCH

A. Pipeline Families

PipeRCA-Bench includes three pipeline families of increasing structural complexity, all implemented with open-source tools and in-memory DuckDB execution.

Fig. 3. PipeRCA-Bench Pipeline Families (node types color-coded)

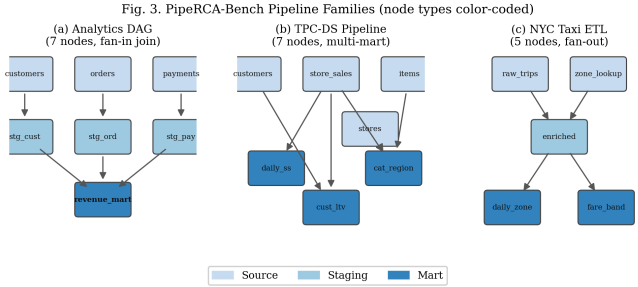

Fig. 3. PipeRCA-Bench pipeline families (node types color-coded).

The **Analytics DAG** is a sales analytics pipeline inspired by the dbt jaffle shop reference project. It consists of three sources, three staging models, and a customer revenue mart in a fan-in join structure, creating a candidate enumeration challenge where multiple upstream sources are equidistant from the observed failure.

The **TPC-DS Pipeline** is a warehouse-scale analytics DAG with four sources feeding directly into three mart tables via multi-path joins. Its direct source-to-mart topology (no staging intermediates) creates a harder structural disambiguation problem: distance-based methods perform especially poorly here because all sources are equidistant from all marts.

The **NYC Taxi ETL** pipeline ingests taxi trip records and a zone lookup table, enriches through a spatial join, and materializes daily zone and fare-band aggregates. This pipeline is also used for the real public-data case study.

B. Fault Taxonomy

Six fault families are defined, grounded in empirical literature [1][15][16]. **Schema drift** captures structural changes that break downstream contracts. **Stale source** captures freshness degradation. **Duplicate ingestion** captures inadvertent row re-ingestion causing aggregate inflation. **Missing partition** captures incomplete delivery of time-partitioned sources. **Null explosion** captures upstream nullification propagating through joins. **Bad join key** captures key mismatches causing join sparsity.

C. Observability Conditions

Three observability conditions test robustness to incomplete runtime lineage. In the **full** condition, runtime lineage matches design lineage exactly. In the **runtime missing root** condition, outgoing lineage edges from the true root are absent, modeling root-cause nodes that failed to emit lineage events. In the **runtime sparse** condition, 30% of non-root edges are randomly dropped, modeling partial instrumentation or event loss.

D. Incident Generation

For each of the 3 × 6 = 18 pipeline-fault combinations, 25 incidents are generated per observability condition, yielding 450 total. Root nodes are sampled from fault-appropriate subsets. Evidence signals are generated stochastically to reflect the fault type, with root nodes receiving elevated run-anomaly and recent-change scores, stale-source roots receiving high freshness-severity, and schema-drift roots receiving contract violations. Decoy nodes receive moderate false-positive signals to model realistic noise.

VI. EXPERIMENTAL EVALUATION

A. Baselines

We compare LineageRank against seven baselines: **Runtime distance** (inverse shortest path in runtime graph); **Design distance** (inverse shortest path in design graph); **Centrality** (blast radius and design support combination); **Freshness only** (freshness severity alone); **Failed tests** (local test failures plus contract violations); **Recent change** (recent-change signal alone); and **Quality only** (evidence-only fusion of quality, freshness, and run-anomaly signals without lineage features).

B. Evaluation Metrics

Primary metrics are Top-1, Top-3, Top-5, MRR, and nDCG. The operational metric is average assets inspected before the true cause (rank – 1). Statistical rigor is provided by bootstrap 95% confidence intervals (1,500 samples) and paired bootstrap significance tests for all primary comparisons.

C. Main Benchmark Results

Table I reports overall performance on PipeRCA-Bench across all 450 incidents. Distance-only baselines produce Top-1 = 0.000 because the root cause is always at distance ≥ 1 from the observed failure. Evidence-only approaches achieve Top-1 0.31 at best. Our four proposed methods substantially outperform all baselines.

TABLE I

Overall RCA Performance on PipeRCA-Bench (450 Incidents)

Method	Top-1	Top-3	MRR	nDCG	Avg. Assets
Runtime distance	0.000	0.318	0.278	0.451	3.262
Design distance	0.000	0.544	0.331	0.495	2.573
Centrality	0.291	0.891	0.573	0.680	1.280
Freshness only	0.167	0.682	0.462	0.595	1.887
Failed tests	0.091	0.640	0.397	0.545	2.224
Recent change	0.224	1.000	0.549	0.664	1.153
Quality only	0.309	0.822	0.558	0.668	1.416
LR-CP (Causal Prop.)	**0.616**	**0.984**	**0.786**	**0.841**	**0.522**
LR-BS (Blind-Spot)	**0.858**	**0.998**	**0.924**	**0.944**	**0.173**
LR-H (Heuristic)	**0.618**	**0.991**	**0.791**	**0.845**	**0.496**
LR-L (Learned RF)	**0.893**	**0.987**	**0.941**	**0.956**	**0.162**

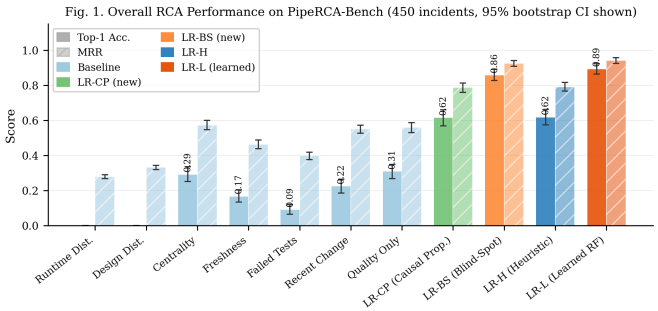

Fig. 1. Overall Top-1 and MRR with 95% bootstrap CIs.

Three findings stand out. First, pipeline RCA is not solved by proximity alone: distance baselines score Top-1 = 0 because they rank candidates by graph distance from v_obs, placing the root cause, which is always further away than v_obs itself, behind closer non-root nodes. Second, evidence-only approaches are insufficient for consistent culprit identification: while individual signals such as freshness can perfectly identify stale-source faults, they fail badly on other fault types. Third, the fused methods substantially outperform all baselines. Notably, **LR-BS achieves Top-1 0.858—a 0.24 absolute improvement over LR-H**—through a principled

multiplicative amplification of the partial-observability signal, achieving near-learned-model accuracy while remaining fully interpretable. LR-CP (0.616) independently validates the causal propagation approach, confirming that directional evidence gradients carry discriminative information beyond what LR-H's additive formulation captures in full-observability conditions.

D. Confidence Intervals and Significance Tests

Table II reports bootstrap 95% confidence intervals for the six key methods. The intervals for LR-BS and LR-L are well separated from baselines and from LR-H.

TABLE II

Bootstrap 95% Confidence Intervals for Key Methods

Method	Top-1 Mean	Top-1 95% CI	MRR Mean	MRR 95% CI
Centrality	0.291	[0.251, 0.336]	0.573	[0.546, 0.599]
Quality only	0.309	[0.267, 0.349]	0.558	[0.530, 0.586]
LR-CP	0.616	[0.571, 0.660]	0.786	[0.760, 0.812]
LR-H	0.618	[0.573, 0.662]	0.791	[0.766, 0.815]
LR-BS	0.858	[0.829, 0.887]	0.924	[0.906, 0.941]
LR-L	0.893	[0.864, 0.920]	0.941	[0.924, 0.957]

Table III reports paired bootstrap significance tests for key comparisons. All differences remain highly significant ($p < 0.001$ under 1,500 bootstrap samples, corresponding to a precision limit of approximately $p < 0.0007$). No multiple-comparison correction was applied across the full method set; the reported comparisons are pre-specified hypothesis tests.

TABLE III

Paired Bootstrap Significance Tests (1,500 Samples, $p < 0.001$ in All Cases)

Comparison	Metric	Mean Diff	95% CI
LR-H vs. Centrality	Top-1	+0.327	[0.271, 0.380]
LR-H vs. Centrality	MRR	+0.218	[0.186, 0.251]
LR-H vs. Centrality	Avg. Assets	–0.784	[–0.916, –0.653]
LR-H vs. Quality only	Top-1	+0.309	[0.251, 0.367]
LR-BS vs. LR-H	Top-1	+0.240	[0.196, 0.284]
LR-BS vs. LR-H	MRR	+0.133	[0.108, 0.158]
LR-BS vs. LR-H	Avg. Assets	–0.322	[–0.403, –0.241]
LR-L vs. LR-H	Top-1	+0.273	[0.224, 0.320]
LR-L vs. LR-H	MRR	+0.149	[0.121, 0.175]

E. Leakage and Ablation Audit

To assess whether LR-L gains reflect genuine learning or exploitation of a single prior, we train four feature-subset variants. Table IV reports the results.

TABLE IV

Leakage and Ablation Audit for LineageRank-L

Feature Variant	Top-1	Top-3	MRR	Avg. Assets
All features	0.893	0.987	0.941	0.162
No fault prior	0.869	0.964	0.922	0.249
Structure only	0.636	0.949	0.779	0.662
Evidence only	0.849	0.989	0.916	0.200

Removing the fault prior reduces Top-1 from 0.893 to 0.869, confirming the model does not succeed primarily through one hand-engineered prior. The evidence-only variant (0.849) outperforms structure-only (0.636), revealing that the benchmark rewards quality and change signals more heavily—a finding we discuss under Threats to Validity.

Table V reports averaged feature importances across held-out pipeline runs.

TABLE V

Top Learned Feature Importances (Averaged Across Held-Out Pipelines)

Feature	Importance	Group
Run anomaly	0.3779	Evidence
Recent change	0.1899	Evidence
Fault prior	0.1033	Prior
Blind spot hint	0.0589	Observability
Uncertainty	0.0522	Observability
Runtime proximity	0.0354	Structure
Runtime support	0.0310	Structure
Blast radius	0.0282	Structure

The model draws on both evidence features (run anomaly, recent change) and partial-observability features (blind-spot hint, uncertainty), confirming that it exploits lineage-coverage signals and not merely quality metrics.

F. Performance by Fault Type

Table VI presents per-fault breakdown for the four proposed methods. Quality only is included as the strongest evidence-only baseline for reference.

TABLE VI

Top-1 Accuracy by Fault Type (450 Incidents, 75 per Fault)

Method	Schema Drift	Stale Src	Dup. Ingest	Missing Part.	Null Expl.	Bad Join
Quality only	0.520	0.640	0.160	0.053	0.133	0.347
LR-CP	0.600	0.760	0.600	0.653	0.400	0.680
LR-BS	0.880	0.947	0.800	0.827	0.813	0.840
LR-H	0.560	0.760	0.627	0.667	0.373	0.720
LR-L	0.867	0.947	0.840	0.933	0.880	0.893

Null explosion is the hardest fault for all heuristics (LR-H: 0.373), because the propagation of null values through joins makes the impacted node set large and evidence signals less discriminative. LR-BS (0.813 on null explosion) substantially closes this gap by exploiting observability discordance even when evidence signals are ambiguous. LR-CP (0.400 on null explosion) confirms that the causal propagation formulation is also harder on fault types with diffuse propagation.

G. Performance by Observability Condition and Pipeline Family

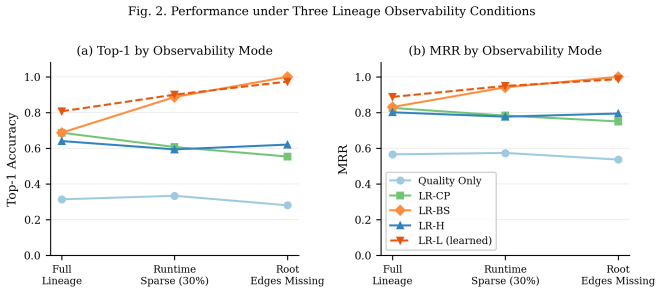

Fig. 2. Performance by observability condition for five methods.

Table VII reveals a striking interaction between observability condition and method. LR-BS achieves **Top-1 = 1.000 under runtime missing root**, perfectly identifying the root cause in all 150 incidents where the root's lineage edges are absent. By contrast, LR-H achieves only 0.620 in this condition—only marginally better than its full-observability performance (0.640). This confirms that LR-H's additive blind-spot feature (weight 0.12) is insufficient to fully exploit the observability gap signal, while LR-BS's multiplicative amplification directly converts this gap into a reliable discriminator.

TABLE VII

Top-1 Accuracy by Observability Condition

Method	Full	Runtime Sparse	Root Edges Missing
Quality only	0.313	0.333	0.280
LR-CP	0.687	0.607	0.553
LR-BS	0.687	0.887	**1.000**
LR-H	0.640	0.593	0.620
LR-L	0.807	0.900	0.973

A secondary finding from Table VII is that LR-L performs better under runtime_missing_root (0.973) than under full observability (0.807). This reflects the Random Forest learning to weight the blind-spot feature heavily when root lineage is absent—effectively learning a similar strategy to LR-BS but from labeled data.

Table VIII reports per-pipeline Top-1 accuracy. LR-H's weakness on tpcds_pipeline (0.267) is notable: the TPC-DS pipeline has a flat structure with no intermediate staging nodes, making proximity-based discrimination harder. LR-BS substantially closes this gap (0.707), while LR-L performs consistently across all three families.

TABLE VIII

Top-1 Accuracy by Pipeline Family

Method	Analytics DAG	TPC-DS Pipeline	NYC Taxi ETL
Centrality	0.247	0.260	0.367
Quality only	0.253	0.407	0.267
LR-CP	0.613	0.627	0.607
LR-BS	0.907	0.707	0.960
LR-H	0.787	0.267	0.800
LR-L	0.900	0.860	0.920

VII. REAL PUBLIC-DATA CASE STUDY

To supplement the controlled synthetic benchmark, we ran a pipeline built on official NYC TLC January 2024 yellow taxi trip records [21] and the official taxi zone lookup CSV. The pipeline ingests 200,000 real rows, enriches through a location join, and materializes two aggregate tables. During execution, runtime lineage events are captured from actual ETL execution. Table IX summarizes pipeline statistics.

TABLE IX

Real NYC Taxi Pipeline Run Statistics

Metric	Value
Dataset mode	Real (TLC January 2024)
Rows ingested	200,000
Distinct pickup zones	235
Daily zone rollup rows	653
Fare band rollup rows	23
Runtime lineage events captured	3

We then constructed a rerun-based case study in which five fault types are injected into 30 separate pipeline runs, with runtime lineage captured from actual execution and evidence signals derived from row-count deltas versus a clean baseline. Table X reports the results.

TABLE X

Real Public-Data Case Study: RCA Performance (30 Incidents)

Method	Top-1	Top-3	MRR	Avg. Assets
Centrality	0.800	1.000	0.900	0.200
Quality only	0.400	1.000	0.667	0.800
LR-H	1.000	1.000	1.000	0.000

LR-H achieves perfect Top-1 across all 30 faulted reruns. This result is stronger than the synthetic benchmark and should be interpreted cautiously: the case study is smaller (30 vs. 450 incidents), structurally simpler (one pipeline family), and covers only five fault types. Its primary value is to demonstrate that the proposed fused RCA framework is not limited to manifest-only synthetic evaluation and can be exercised on actual public data with captured runtime lineage events.

VIII. DISCUSSION

A. Novelty and Positioning

The novelty of this paper lies not in the general idea that lineage helps debugging—provenance and lineage systems have long supported inspection and dashboard-driven tracing—but in three specific contributions. First, we formulate pipeline RCA as a ranked retrieval problem with a reproducible benchmark, enabling systematic comparison of methods across fault types, pipeline families, and observability conditions. Second, we introduce LR-BS, which demonstrates that design-runtime lineage discordance is a principled and powerful discriminative signal for the partial-observability condition, achieving near-learned-model accuracy through a simple multiplicative rule. Third, we show that the observability condition itself fundamentally changes the relative ranking of methods: LR-BS is optimal under runtime-missing-root, LR-CP generalizes better under full observability than LR-H on certain pipelines, and the learned model adapts across all conditions.

This positioning distinguishes our work from three adjacent lines. Provenance systems [12][13][14] focus on collecting and querying derivation metadata without benchmarked ranked RCA. Observability tools support browsing and alerting but not systematic method comparison. RCA benchmarks in adjacent domains [11][19] operate over different graph semantics, evidence types, and failure modes than those found in data pipelines; PC-PageRank achieves 9% Top-1 on RCAEval while our strongest heuristic achieves 86% on PipeRCA-Bench, which underscores that direct transfer of methods across domains is not appropriate.

B. Threats to Validity

The first threat is **incident realism**. PipeRCA-Bench uses controlled synthetic incidents rather than organically collected production failures. We mitigate this by grounding the fault taxonomy in empirical literature [1][15][16], including multiple pipeline families, and providing a real-data validation layer.

The second threat is **benchmark bias toward evidence features**. The leakage and ablation audit shows that evidence-only learning achieves Top-1 0.849 while structure-only achieves 0.636. This reflects the synthetic signal generation concentrating discriminative evidence at root nodes. We emphasize LR-H and LR-BS as operational contributions and treat LR-L as an upper-bound learned variant.

The third threat is **external validity across tool ecosystems**. The benchmark is implemented with OSS-first assumptions. Transfer to proprietary warehouses or heterogeneous observability stacks may require adaptation.

The fourth threat is **ground-truth simplification**. Each incident has one designated root cause. Real incidents may involve cascading or

compound causes. We treat the single-primary-cause formulation as a tractable starting point.

The fifth threat is **case-study scale**. The real public-data validation covers 30 incidents on a single pipeline family and achieves a perfect score, which reflects both the method's robustness and the relative simplicity of the setting.

IX. CONCLUSION

This paper introduces PipeRCA-Bench and four LineageRank variants for root-cause ranking in data pipelines under partial observability. We formulate pipeline RCA as ranked retrieval, present a 450-incident reproducible benchmark, and show that fusing lineage, causal propagation, and observability-coverage signals substantially outperforms single-signal baselines. LR-BS, our novel partial-observability-aware heuristic, achieves Top-1 0.858—a 0.24 absolute improvement over the additive heuristic LR-H and within 0.04 of the Random Forest—while remaining fully interpretable. Under runtime-missing-root conditions it achieves perfect Top-1 1.000 by directly exploiting design-runtime lineage discordance as a root-cause indicator. Bootstrap confidence intervals and paired significance tests confirm that all headline gains are not marginal. A leakage audit confirms the learned model exploits genuine signals, and a real public-data case study on NYC TLC records validates realism.

Future work includes expanding PipeRCA-Bench with additional pipeline families and compound fault incidents, adding a temporal train-test split for LR-L, and studying how LR-BS generalizes to production observability stacks with noisier or more heterogeneous lineage coverage.

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